Supplementary Numerical Displays: When Can Markets Crash Without News?

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Scope of the supplement

These panels and the earlier nonoptimal two-day witness are preserved for replication and illustration. They are not independent estimates of real market risk. All mathematical claims and the improved interval-certified construction appear in the main manuscript.

Earlier exploratory two-day execution

The preceding theorems form a hierarchy of *necessary* tests. Passing the rolling-band bound, the scalar selling-rate test, the dollar-closure preconditioning test, and the cut-capacity test still does not prove a path exists. We therefore turn from exclusion and lower bounds to a constructive example that enforces the constraints *simultaneously*. To test repeated *partial-coverage* lower-limit closes, we constructed a continuous $N = 10$, $k = 9$ example in the $g(S) \equiv \eta$ market. The nine targets use symmetric $\pm 3\%$ proportional bands; the compensator uses a wider symmetric $\pm 25\%$ band. These are *hypothetical normalized assets*, not calibrated TSE stocks. Set $F = \eta = 1$, $\rho = 0.4$, equal session lengths $\tau_1 = \tau_2 = 20$, and a common gross share-rate limit $\bar{v} = 1$.

Write $S_1 = \cdots = S_9 = x(t)$ and $S_{10} = y(t)$. On each session let $x(t)$ be a continuous piecewise-linear path: it rises to a first-session plateau h_1 then closes at 0.97, rises to a second-session plateau $h_2 = 0.995$ then closes at $0.97^2 = 0.9409$. All ramps last 0.7 trading-time units. Dollar pairing uniquely determines the compensator's ODE via

$$u(t) = x(t)[\dot{x}(t) + \rho(x(t) - 1)], \qquad \dot{y} = -\rho(y - 1) - \frac{9u(t)}{y(t)}, \quad y(0) = 1. \tag{1}$$

The first plateau $h_1 \approx 1.0142982125$ is selected from the *exact piecewise-polynomial* equation

$$X_i(T_2) = \frac{x(T_2)^2 - 1}{2} + \rho \int_0^{T_2} x(t)[x(t) - 1]\,\mathrm{d}t = 0, \quad i = 1, \ldots, 9. \tag{2}$$

Because $\dot{X}_{10} = -9\dot{X}_1$, the compensator's terminal executed-dollar balance closes as well. Independent high-accuracy ODE and Runge–Kutta integrations agree for $y(T_2)$ to approximately 2×10^{-12}; band and rate slack were checked on a dense time grid. These numerical checks are an explicit *witness*, not a proof of general N-asset sufficiency or a certified interval-enclosure theorem.

The example proves neither a trader's incentive to undertake the sequence nor a *locked* limit-down order book. It does show numerically that rolling bands, finite executed-share rates, no fundamental-news shocks, and terminal dollar closure can be compatible with *multiple* lower-limit closes in a broad subset of assets. For a crisis threshold, the breadth requirement alone does not suffice: for this path $Q_{0.9}(T_2) = 0.060918$ but the average-loss restriction binds, giving $\mathcal{C}_{0.9}(T_2) = 0.019932$ (thus, for example, $\delta = 0.015$ is reached). No claim of an exact minimum rate, minimum turnover, or equilibrium occurrence follows.

Other diagnostic displays

Table 1: Two-session $N = 10$ deterministic closing-limit witness (normalized units). Day 1 and Day 2 target losses are 3% each relative to the reset reference; the terminal systemic functional is smaller because the compensator rises.

Quantity	Verified value
Nine target closing prices, session 1 / session 2	0.970000 / 0.940900
Compensator closing prices, session 1 / session 2	1.228104 / 1.417574
Maximum individual gross executed-share rate	$0.680780 < \bar{v} = 1$
Target executed-dollar balance, day 1 / day 2	$+0.078235$ / -0.078235
Maximum absolute terminal net-dollar balance	$< 7 \times 10^{-15}$
Total paired-dollar turnover / largest gross share sales	2.59310 / 1.46241
Terminal $\mathcal{C}_{0.9}(T_2)$ / breadth loss $Q_{0.9}(T_2)$	0.019932 / 0.060918

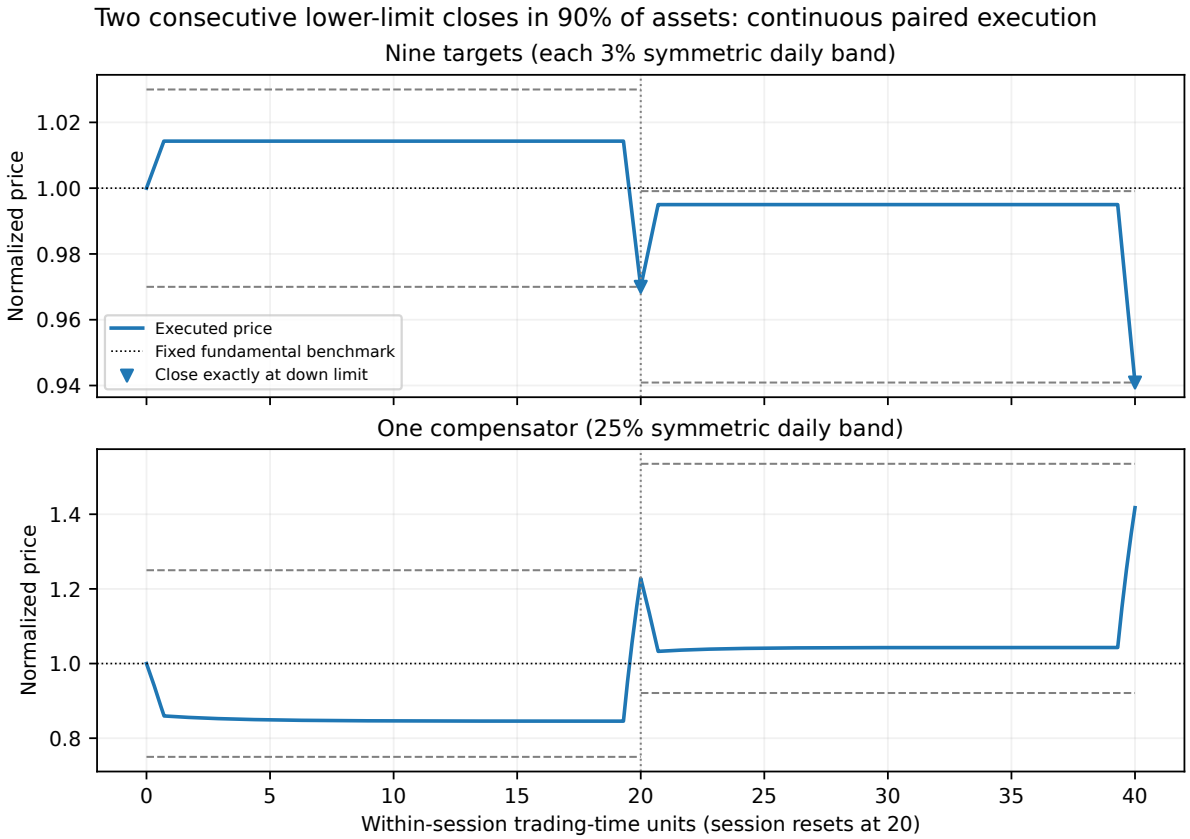


Figure 1: Repeated partial-coverage lower-limit closing prices under rolling daily bands. The same nine target assets close at their 3% lower limits twice. The tenth, higher-band compensating asset stays within its own 25% limits. The displayed paths contain no instantaneous trades. The horizon and bands are specified model inputs, not observations.

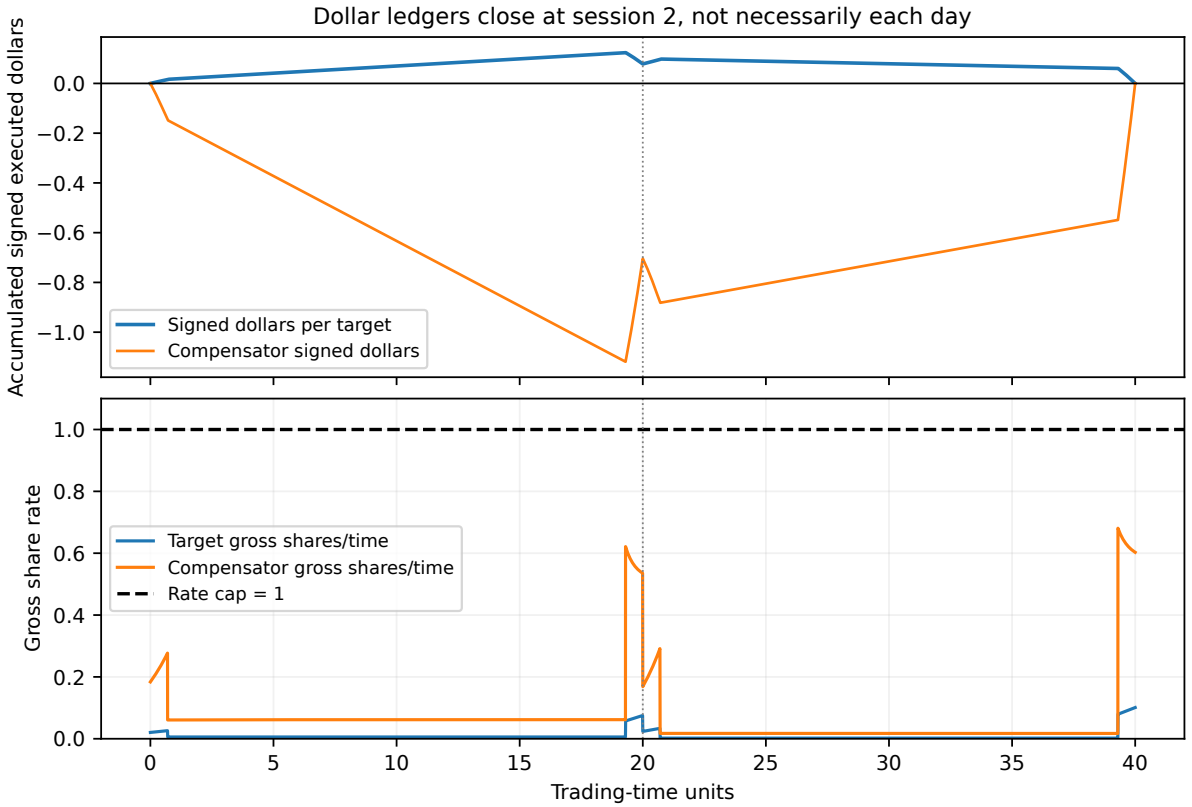


Figure 2: Trading ledgers and gross-rate check for the same two-session witness. The first session builds positive net dollar flow in the targets; the second reverses it, leaving every asset's final signed-dollar balance zero. Gross rates remain below the imposed share-rate limit; dollar balance is enforced continuously, not just at maturity.

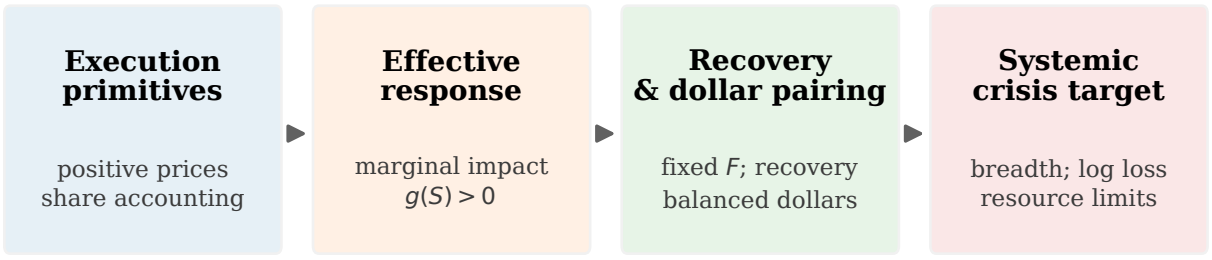


Figure 3: One explicitly defined ideal market. Price priority, signed-share accounting, effective price impact, resilience, and paired-dollar trading have distinct mathematical roles. All quantitative panels in this paper are illustrations of this specified model rather than observations of a live order book.

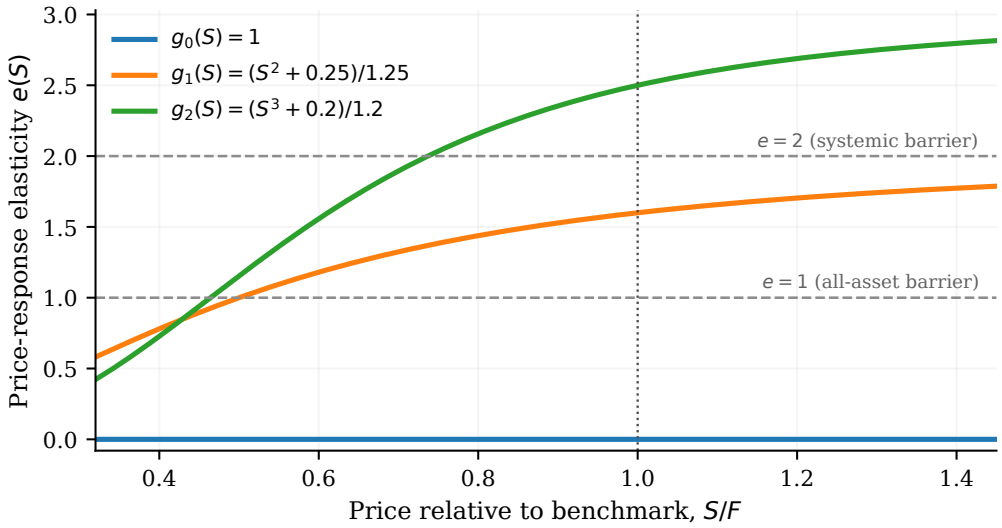


Figure 4: Local elasticity of three positive, smooth, non-power-law impact functions. The dotted vertical line marks the common initial price $S = F$. The global elasticity-barrier theorems in the main manuscript require their elasticity inequalities throughout the *relevant attainable price region*; a value at F alone gives no global classification. This is a formula-based illustration, not empirical market depth.

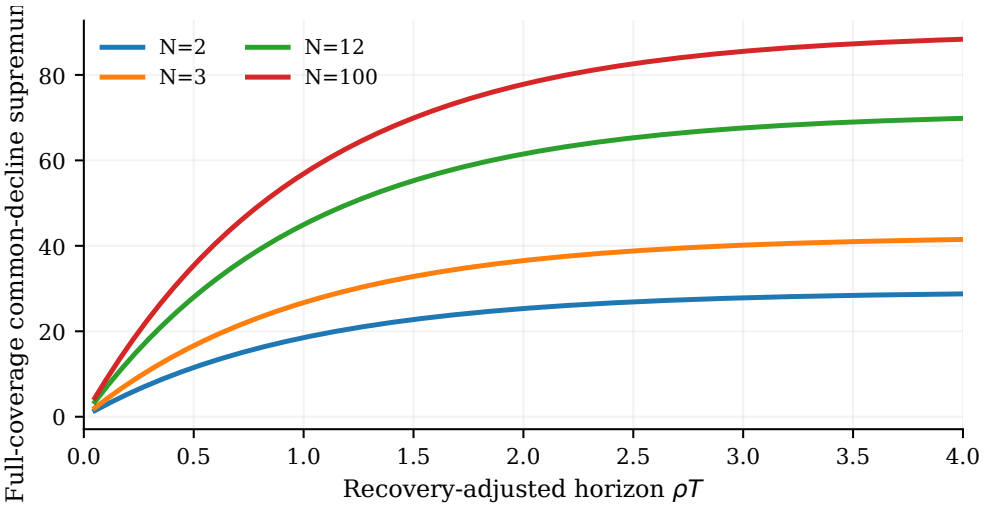


Figure 5: A sharp, model-specific common-decline boundary. For $g = \eta$ and $k = N$, the upper envelope of the common simple-price decline is $100(1 - e^{-\delta_{\mathrm{crit}}})\%$. Every strictly smaller log-loss threshold is reachable with some finite budget; the boundary itself is not attained with strictly positive prices. This is not a probability curve.

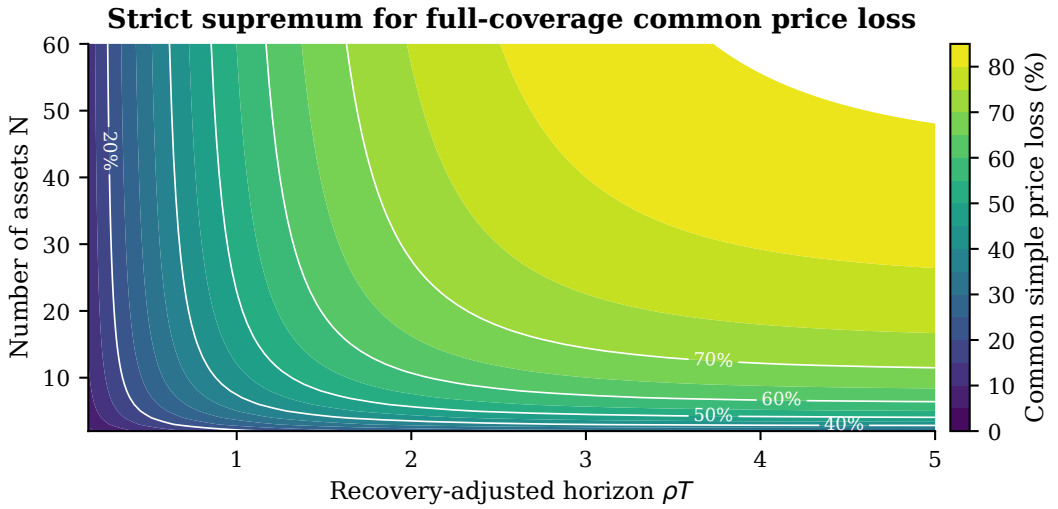


Figure 6: The full-coverage boundary across asset count and recovery time. Color reports $100(1 - e^{-\delta_{\mathrm{crit}}(N,T)})$, the supremum of simultaneous *simple-price* losses for constant per-share impact with individually closed terminal dollar flows. No probability, observed drawdown, or finite-budget optimum is inferred.

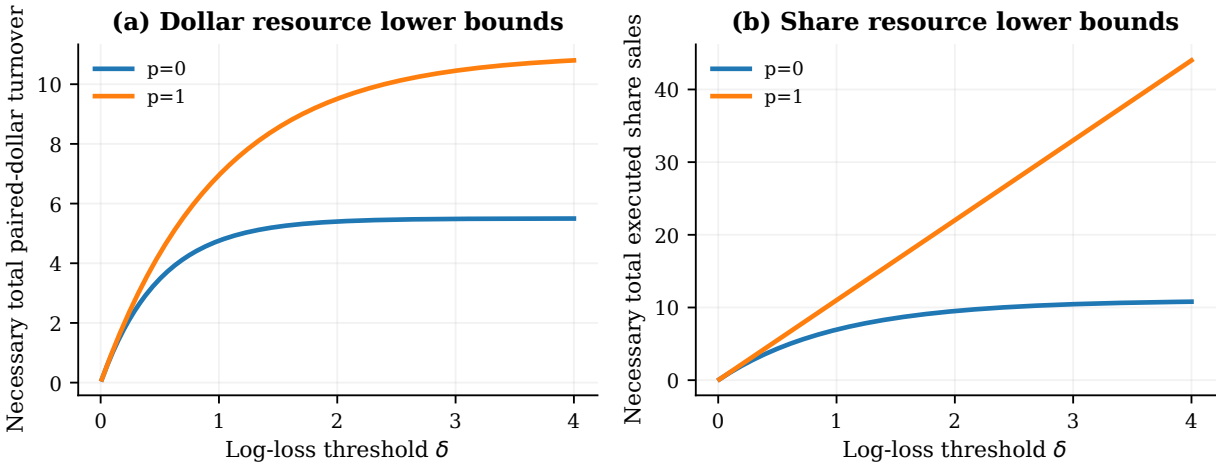


Figure 7: General necessary resource floors illustrated for the power-response family. With $F = \eta = 1$ and $k = 11$, the curves evaluate $k \int_{e^{-\delta}}^{1} S/g(S)\,\mathrm{d}S$ (dollars) and $k \int_{e^{-\delta}}^{1} 1/g(S)\,\mathrm{d}S$ (shares), for $p = 0, 1$. The bounds are *not* the optimal execution cost and do not establish reachability.

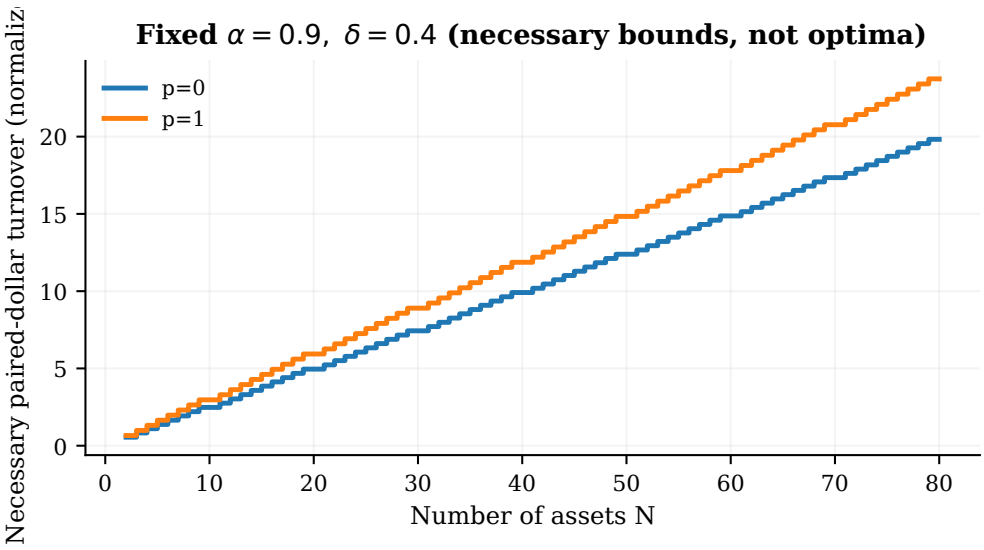


Figure 8: Dependence on asset number of the necessary dollar-resource bound. The experiment copies per-asset benchmark price and response parameters, fixes $\alpha = 0.9$ and $\delta = 0.4$, and increases the number of assets N. The steps arise from the discrete coverage count $\lceil \alpha N \rceil$. It is not a capitalization-scaled prediction for an actual stock index.

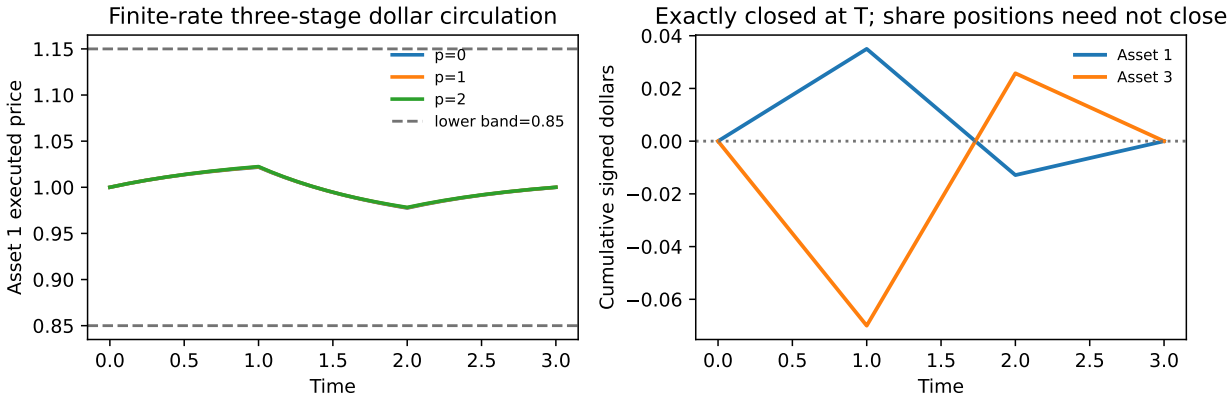


Figure 9: A new bounded-rate, terminally dollar-closed path (not a pulse approximation). For $N = 3$, $T = 3$, $a = .035$, $F = \rho = \eta = 1$, and $u = (1, 1, -2)$, the price trajectories under $p = 0, 1, 2$ share the same absolutely continuous dollar-rate signal; right panel shows dollar ledgers that close exactly. The peak per-asset gross share rate is below 0.15, and all prices remain inside a $\pm 15\%$ band.

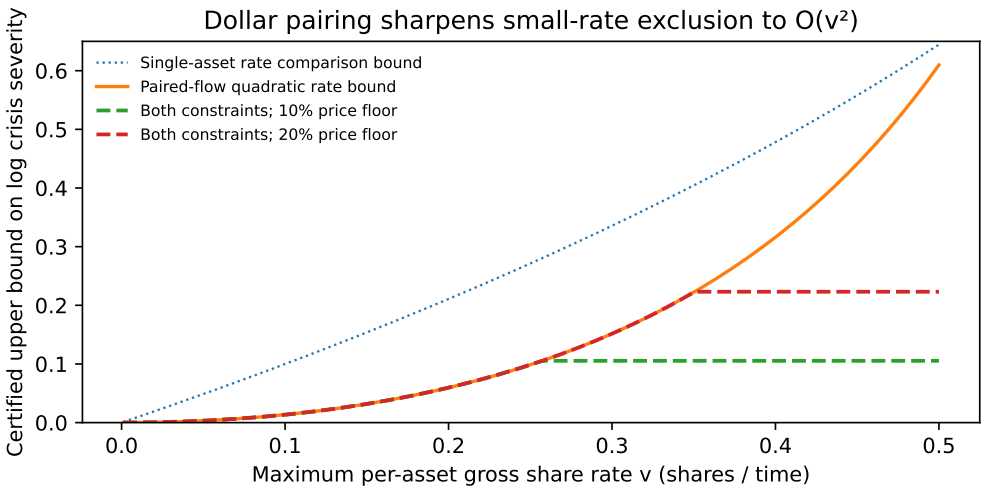


Figure 10: Exclusion certificates, not exact reachable maxima. For $g = \eta = F = \rho = 1$ and $T = 3$, a paired-flow quadratic estimate improves the single-price comparison at small rates; a single-session lower price band provides an additional hard ceiling. Neither curve reports optimal crisis control or an observed market frequency.

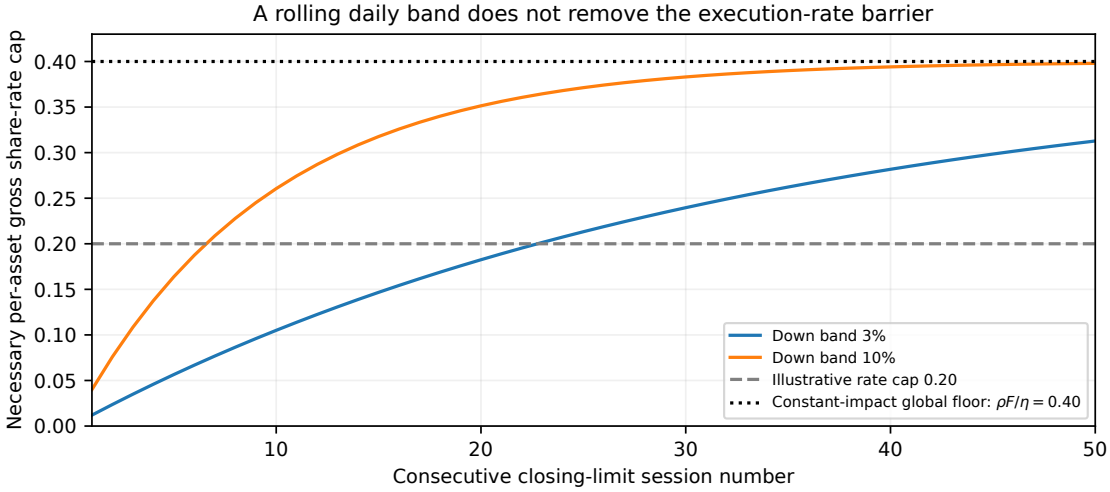


Figure 11: Necessary gross share-rate floor for repeated lower-limit closes. The scalar constant-impact necessary-rate formula in the main manuscript in the scalar constant-impact comparator, $F = \eta = 1$, $\rho = 0.4$, $\tau = 20$. These are *necessary* per-asset rates; paired execution, available inventory, and the compensator's own daily band can make the actual market threshold higher.

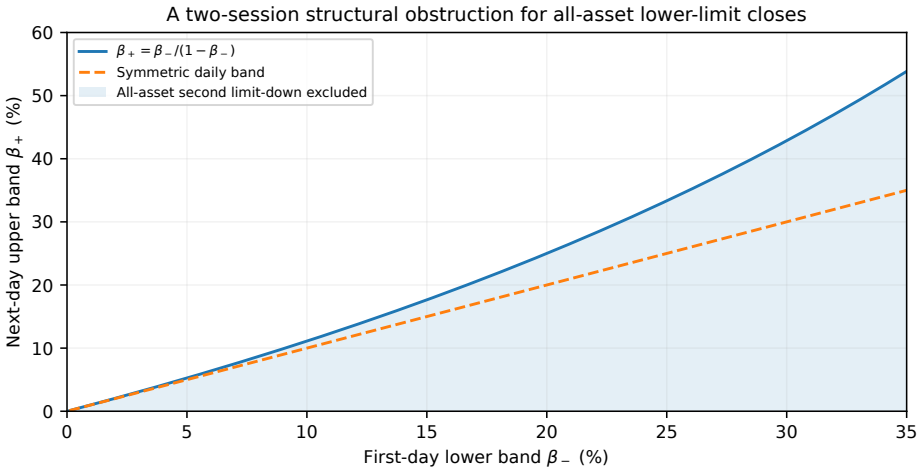


Figure 12: Parameter region in which the second all-asset lower-limit close is ruled out. The region beneath the curve satisfies $(1+\beta_+)(1-\beta_-) \leq 1$. Symmetric proportional bands lie strictly inside this region. This is a conditional invariant-region result, not a model of exchange-wide order queues.

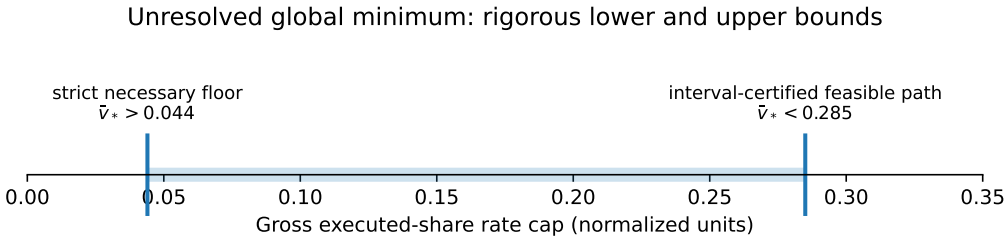


Figure 13: What is and is not optimized. A strict rate floor from the existing quadratic systemic bound and a certified feasible execution path bracket the unknown global minimum $\bar{v}_*$. The interval is a logical admissibility bracket, not a statistical confidence interval.

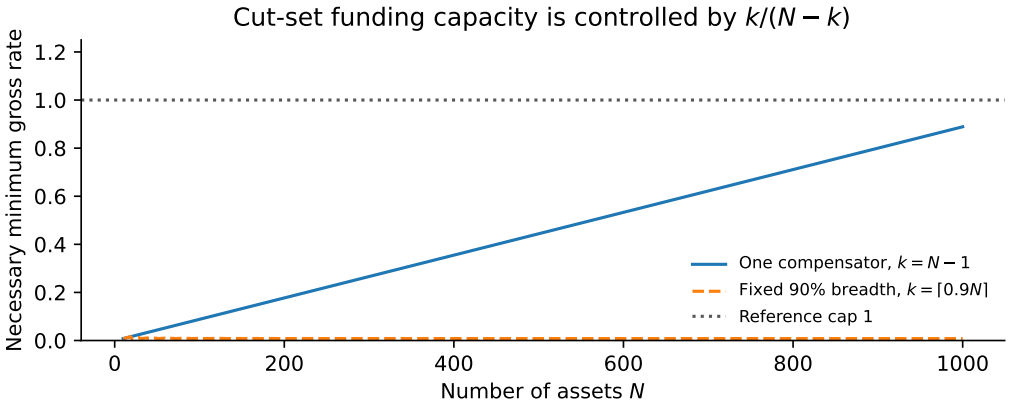


Figure 14: A funding-capacity constraint that grows with the falling majority. The necessary second-day rate in the main manuscript increases as $k/(N-k)$ when a single asset absorbs the net matched dollars of an expanding majority. This does not prove that a feasible market exists at every plotted N.

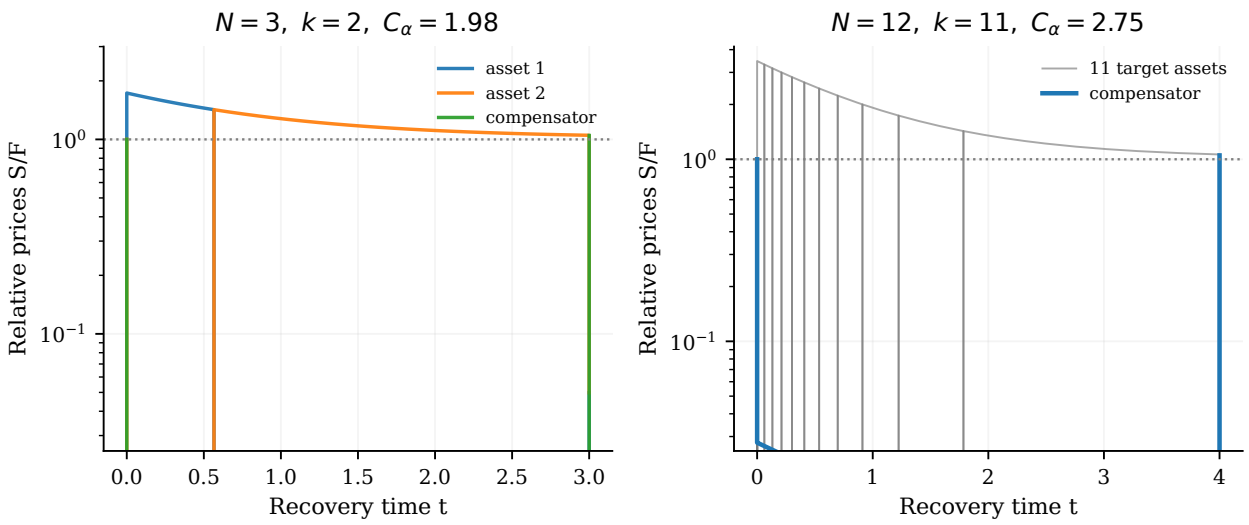


Figure 15: Constructed price paths with terminally closed dollar flows. For $N = 3$ and $N = 12$, one compensating asset ends above its initial value, while the other $N - 1$ prices fall by the common target log loss $L = 3$. Logarithmic vertical scales keep near-zero prices visible. Kinks and jumps are permitted control pulses, not simulated exchange events.

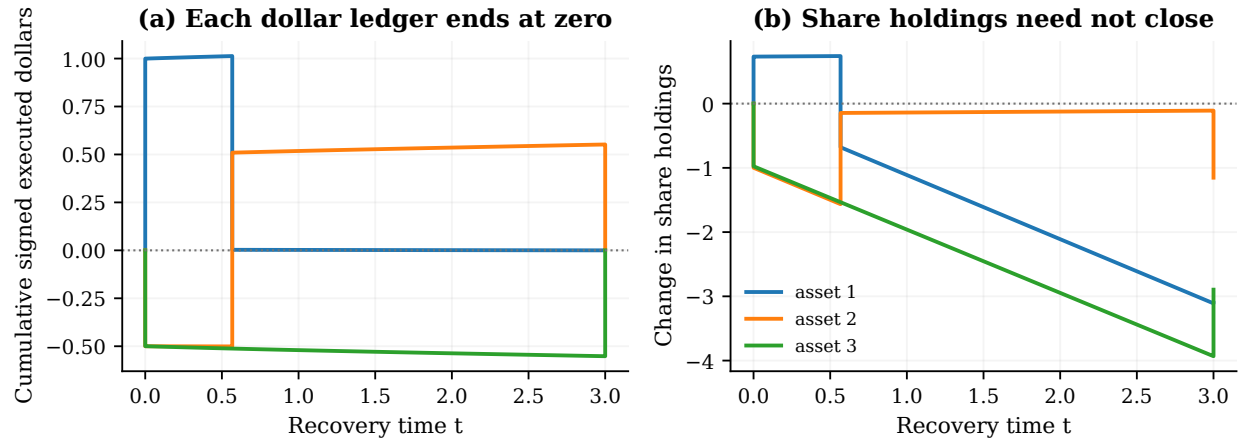


Figure 16: Net executed dollars are not net shares. A separate $N = 3$, $k = 2$, $T = 3$, $L = 5$ witness has $X_i(T) = 0$ for every asset (left), but its net share positions generally differ from zero (right). Initial share inventories must cover gross sales; transaction closure alone does not supply them.

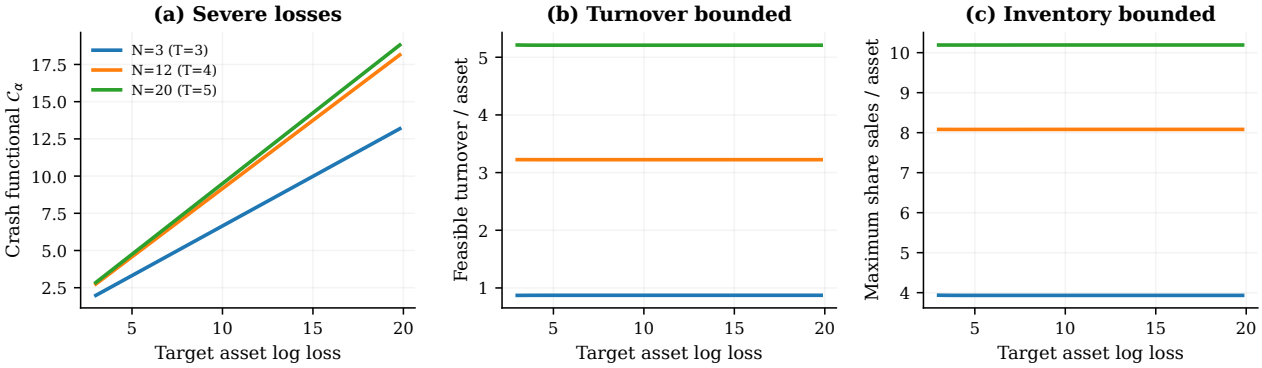


Figure 17: Increasing crisis severity with bounded *feasible* resources. Selected $N = 3, 12, 20$ construction families use $k = N - 1$ and horizons $T = 3, 4, 5$, respectively. As the terminal target-price log loss increases, achieved $\mathcal{C}_\alpha$ rises (left), whereas the particular paths' turnover per asset (center) and largest cumulative share sale (right) remain bounded. The different horizons and prices are model choices, not a cross-market calibration or an optimum.

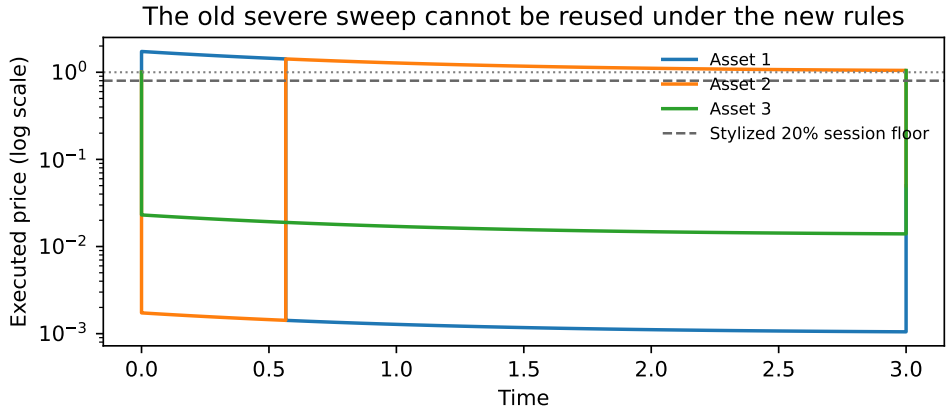


Figure 18: Diagnostic rejection of an old severe path under new execution restrictions. The $N = 3$, $p = 0$ severe-dollar-closed witness from the unrestricted model crosses a stylized one-session 20% lower band and includes nonzero instantaneous pulses. Its exact closed dollar ledger therefore *does not* certify reachability under either a finite gross rate or the price-limit protocol.